

CBSE Class 12
Computer Science
Previous Year Question Paper 2013

Series: SKS/1

Code no. 91/1

- Please check that this question paper contains **15** printed pages.
- Code number given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- Please check that this question paper contains **7** questions.
- **Please write down the Serial Number of the question before attempting it.**
- 15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.

COMPUTER SCIENCE

Time Allowed: 3 hours

Maximum Marks: 70

Instructions:

- All questions are compulsory.**
- Programming Language: C++**

1. (a) What is the benefit of using function prototype for a function? Give a suitable example to illustrate it using C++ code. 2 Marks

Ans: A function prototype is used to inform the compiler of a function declared somewhere in the program later after its call is made in the program.

Example: void func(); //function protoype

```
void main(){
    func();
}

void func(){
    cout<<"Function declaration.";
}
```

(b) Observe the following C++ code and write the name(s) of the header file(s), which will be essentially required to run it in a C++ compiler: 1 Mark

```
void main( )
{
    int Number;
    cin>>Number;
    if (abs(Number) ==Number);
        cout<<"Positive"<<endl ;
}
```

Ans: math.h and iostream.h are the required header files for given code.

(c) Observe the following C++ code carefully and rewrite the same after removing all the syntax error(s) present in the code. Ensure that you underline each correction in the code. 2 Marks

Important Note:

- All the desired header files are already included, which are required to run the code.
- Correction should not change the logic of the program.

```
#define Convert (P,Q) P+2*Q;

void main ()
{
    Float A, B, Result;
    cin>>A>>B;
    Result=Convert [A,B]
    cout<<"Output:"<<Result<<endl;
}
```

Ans: #define Convert (P,Q) P+2*Q;

```
void main ()
{
    float A, B, Result;
    cin>>A>>B;
    Result=Convert(A,B);
    cout<<"Output : "<<Result<<endl;
}
```

(d) Observe the following C++ code carefully and obtain the output, which will appear on the screen after execution of it. 3 Marks

Important Note :

- All the desired header files are already included in the code, which are required to run the code.

```
void main ( )
{
    char *String= "SHAKTI";
    int *Point, Value[]={10,15,70,19};
```

```
Point=Value;
cout<<*Point<<String<<endl;
String++;
Point++;
cout<<*Point<<String<<endl;
}
```

Ans: The output for the given program is as follows:

10SHAKTI

15HAKTI

(e) Observe the following C++ code carefully and obtain the output, which will appear on the screen after execution of it. 3 Marks

```
#include<iostream.h>

class Aroundus
{
    int Place, Humidity, Temp;
public:
    Aroundus(int P=2) {Place=P; Humidity=60; Temp=20;}
    void Hot(int T) {Temp+=T;}
    void Humid(int H) {Humidity+=H;}
    void JustSee()
    {
        cout<<Place<<":"<<Temp <<"&"<<Humidity<<"%"<<endl;
    }
};

void main ( )
```

```
{
    Aroundus A,B(5);
    A.Hot(10);
    A.JustSee();
    B.Humid(15);
    B.Hot(2);
    B.JustSee();
    A.Humid(5);
    A.JustSee();
}
```

Ans: The following is the output for the given program:

2:30&60%

5:22&75%

2:30&65%

(f) Based on the following C++ code, find out the expected correct output(s) from the option (i) to (iv). Also, find out the minimum and the maximum value that can be assigned to the variable Trick used in the code , at the time when value of Count is 3. 2 Marks

```
void main()
{
    char Status[][10]={"EXCEL","GOOD", "OK"};
    int Turn=10,Trick;
    for(int Count=1;Count<4;Count++)
    {
        Trick=random(Count);
    }
}
```

```
cout<<Turn-Trick <<Status[Trick] <<"#";
    }
}
```

- (i) 10EXCEL#10EXCEL#8OK#
- (ii) 10EXCEL#8OK#9GOOD#
- (iii) 10EXCEL#9GOOD#10EXCEL#
- (iv) 10EXCEL#10GOOD#8OK#

Ans: (i), (iii) are the expected outputs of the given code.

Maximum Value of Trick = 0

Minimum Value of Trick = 2

2. (a) Write any two difference between Constructor and Destructor. Write the function headers for constructor and destructor of a class Member.

2 Marks

Ans: The differences between the Constructors and Destructors of a class are:

Constructors	Destructors
Constructors are used to initialise the data members of the class.	Destructors are used to free the memory occupied by the object of the class.
Constructors of a class can be overloaded.	Destructors can't be overloaded. Each class will have only one destructor.

Example: class testexample{

public:

testexample(); //constructor

~testexample(); //destructor

};

(b) Answer the questions (i) and (ii) after going through the following class:
2 Marks

```
class Motor
{
    int MotorNo, Track ;
    public:
        Motor();    //Function 1
        Motor(int MN);    //Function 2
        Motor(Motor &M) ;    //Function 3
        void Allocate();    //Function 4
        void Move();    //Function 5
};

void main()
{
    Motor M;
    :
    :
}
```

(i) Out of the following, which of the option is correct for calling Function 2 ?

Option 1-Motor N(M);

Option 2-Motor P(10);

Ans: Option 2 is the correct call to Function 2.

(ii) Name the feature of Object Oriented Programming, which is illustrated by Function 1, Function 2 and Function 3 combined together.

Ans: Function overloading or polymorphism is the feature of OOPs illustrated by function 1, 2 and 3.

(c) Define a class Tourist in C++ with the following specifications: 4 Marks

Data Members

- **Carno** – to store Bus No
- **Origin** — to store Place name of origin
- **Destination** – to store Place name of destination
- **Type** – to store Bus Type such as ‘E’ for Economy
- **Distance** – to store the Distance in Kilometres
- **Charge** -to store the Car Fare

Member Functions

- A constructor function to initialize Type as ‘E’ and Freight as 250
- A function **CalcCharge()** to calculate Fare as per the following criteria:

TYPE	FARE
‘E’	16*Distance
‘A’	22*Distance
‘L’	30*Distance

- A function **Enter()** to allow user to enter values for Carno, Origin, Destination, Type and Distance. Also, this function should call **CalcCharge()** to calculate Fare.
- A function **Show()** to display the content of all the data members on screen.

Ans: class Tourist

```
{
    int Carno;
    float Distance, Charge;
```

```
char Origin[20], Destination[20];  
  
char Type;  
  
void CalcCharge(char Type){  
    if(Type == "E")  
        Charge = 16*Distance;  
    else if(Type == "A")  
        Charge = 2*Distance;  
    else if(Type == "L")  
        Charge = 30*Distance;  
}  
  
public:  
    Tourist(){  
        Type = "E";  
        Charge = 250;  
    }  
  
    void Enter(){  
        cout<<"Enter Bus number:";  
        cin>>Carno;  
        cout<<"Enter Place name of origin:";  
        cin>>Origin;  
        cout<<"Enter Place name of destination:";  
        cin>>Destination;  
        cout<<"Enter Distance:";  
        cin>>Distance;
```

```

        cout<<"Enter Type('E','A','L'):";
        cin>>Type;
        CalcCharge(Type);
    }
    void Show(){
        cout<<"Bus number:"<<Carno<<endl;
        cout<<"Origin place name:"<<Origin<<endl;
        cout<<"Destination place name:"<<Destination<<endl;
        cout<<"Distance:"<<Distance<<endl;
        cout<<"Type:"<<Type<<endl;
        cout<<"Charge:"<<Charge<<endl;
    }
};

```

(d) Consider the following C++ code and answer the questions from (i) to (iv): **4 Marks**

```

class Student
{
    int Class,Rno;
    char Section;
protected:
    char SName[20];
public :
    Student();
    void Stentry();
    void Stdisplay();

```

```
};  
  
class Score: private Student{  
    float Marks[5];  
protected:  
    char Grade[5];  
public:  
    Score();  
    void Sentry();  
    void Sdisplay();  
};  
  
class Report: public Score  
{  
    float Total, Avg;  
public:  
    char OverallGrade, Remarks[20];  
    Report();  
    void REvaluate();  
    void RPrint();  
};
```

(i) Which type of Inheritance is shown in the above example ?

Ans: Multilevel Inheritance is used in the above example.

(ii) Write the names of those data members, which can be directly accessed from the objects of class Report.

Ans: OverallGrade and Remarks are the only accessible data members for class Report.

(iii) Write the names of those member functions, which can be directly accessed from the objects of class Report.

Ans: REvaluate(), RPrint(), Sentry(). Sdisplay() are accessible by object of Report class.

(iv) Write the names of those data members, which can be directly accessed from the Sentry() function of class Score.

Ans: SName, Marks, Grade are only members accessible directly by objects of Sentry() function of class Score.

3. (a) Write code for a function void Convert(int T[],int Num) in C++, which re-positions all the elements of the array by shifting each of them to one position before and by shifting the first element to the last position. 3 Marks

For example: If the content of array is

0	1	2	3	4
22	25	70	32	12

The content will be

0	1	2	3	4
25	70	32	12	22

Ans: void Convert(int T[],int Num)
{

int C = T[0];

for(i=0;i<Num-1;i++)

T[i] = T[i+1];

T[Num-1] = C;

cout<<"New Array:";

for(i=0;i<Num;i++)

```
cout<<T[i]<<" ";
}
```

(b) An array P[15][10] is stored along the column in the memory with each element requiring 4 bytes of storage. If the base address of array P is 14000, find out the location of P[8][5]. **3 Marks**

Ans: Lr = 0 , Lc = 0, Base address = 14000, W = 4, R = 15, C = 10

Address for (i,j)th element along the column = Base + W((i-Lr)+R(j-Lc))

Address for (8,5)th element = Base + 4((8)+15(5))

Address = 14000 + 4(8 + 75)

Address = 14000 + 4*83

Address = 14000 + 332

Address = 14332

(c) Write a user defined function DispNTen(int L[][4], int R, int C) in C++ to find and display all the numbers, which are not divisible by 10. For example if the content of array is: **2 Marks**

20	17	30
12	19	10

The output should be

17 12 19

Ans: void DispNTen(int L[][4], int R, int C)
{

cout<<"Elements in array not divisible by 10: ";

for(int i=0;i<R;i++)

for(int j= 0;j<C;j++)

if(L[i][j]%10==0)

cout<<L[i][j]<<" ";

}

(d) Evaluate the following post fix expression. Show the status of stack after execution of each operation: 2 Marks

60, 6, /, 5, 2, *, 5, -, +

Ans: The stack is as follows:

Input	Operation	Stack
60	Push 60	60
6	Push 6	60, 6
/	Pop(2)	
	Push 10	10
5	Push 5	10, 5
2	Push 2	10, 5, 2
*	Pop(2)	10
	Push 10	10, 10
5	Push 5	10, 10, 5
-	Pop(2)	10
	Push 5	10, 5
+	Pop(2)	
	Push 15	15

The list has finished and in stack we only have one element which is the answer after evaluation i.e. 15.

(e) Write a function QINSERT() in C++ to perform insert operation on a Linked Queue, which contains Client no and Client name. Consider the following definition of NODE in the code of QINSERT(). 4 Marks

struct NODE

```
{
    long int Cno;    //Client No
    char Cname[20]; //Client Name
    NODE *Next;
};
```

Ans: void QINSERT(){

```
    NODE *P = new NODE();
    cout<<"Enter the Client Number:";
    cin>>P->Cno;
    cout<<"Enter the Client Name: ";
    cin>>P->Cname;
    P->Next = NULL;
    if(f == NULL && r == NULL)
        f = r = P;
    else{
        r->Next = P;
        r = P;
    }
}
```

4. (a) Fill in the blanks marked as Statement 1 and Statement 2, in the program segment given below with appropriate functions for the required task.

1 Mark

```
class Customer
```

```
{
```

```

long int CNo;                //Customer Number

char CName[20];             //Customer Name

char Email[30];             //Email of Customer

public:

    void Allocate();          //Function to allocate a member

    void Show();              //Function to show customer data

    void ModifyEmail()        //Function to modify Email
    {
        cout<<"Enter Modified Email:";
        gets(Email);
    }

    long int GetCno() {return CNo;}
};

void ChangeData()
{
    fstream File;
    File.open("CUST.DAT",ios::binary|ios::in|ios::out);

    int Change=0, Location;

    long int ChangeCno;

    cout<<"Cno - whose email required to be modified:";
    cin>>ChangeCno;

    Customer CU;

    while(!Modify && File.read((char*)&CU,sizeof(CU)))
    {

```

```

        if (CU.GetCno()==ChangeCno)
        {
            CU.ModifyEmail();
            Location=File.tellg()- sizeof (CU);
            //Statement 1:To place file pointer to the required position
            _____;
            //Statement 2:To write the object CU on to the binary file
            _____;
            Change++;
        }
    }
    if (Change)
        cout<<"Email Modified..."<<endl;
    else
        cout<<"Customer not found..."<<endl;
    File.close();
}

```

Ans: File.seekg(Location); //statement 1

File.write((char*)&CU, sizeof(CU)); //statement 2

(b) Write a function CountHisHer() in C++ which reads the contents of a text file diary.txt and counts the words. His and Her (not case sensitive). 2 Marks

For example, if the file contains:

Pinky has gone to his friend's house. His friends name is Ravya. Her house is 12 KM from here.

The Function should display the OUTPUT as:

Count for His : 2

Count for Her : 1

Ans: void CountHisHer()

```
{  
    ifstream f;  
    f.open("diary.txt");  
    char str[100];  
    int his = 0, her = 0;  
    while(!f.eof())  
    {  
        f getline(str,100,".");  
        if(strcmp(str,"His")==0)  
            his++;  
        if(strcmp(str,"Her")==0)  
            her++;  
    }  
    cout<<"Count for His:"<<his<<endl;  
    cout<<"Count for Her:"<<her<<endl;  
    f.close();  
}
```

(c) Assuming the class VINTAGE as declared below, write a function in C++ to read the objects of VINTAGE from binary file VINTAGE.DAT and display those vintage vehicles, which are priced between 200000 and 250000.

3 Marks

class VINTAGE

```
{
```

```

int VNO;    //Vehicle Number

char VDesc[10]; //Vehicle Description

float Price;

public:

void GET(){cin>>VNO; gets(VDesc); cin>>Price;}

void VIEW()
{
    cout<<VNO<<endl;
    cout<<VDesc<<endl;
    cout<<Price<<endl;
}

float ReturnPrice() {return Price;}
};

Ans: void Show()
{
    VINTAGE Vin;
    ifstream f;
    f.open("VINTAGE.DAT",ios::binary);
    while(f.read((char*)&Vin,sizeof(Vin)))
    {
        if(Vin.ReturnPrice()>200000 && Vin.ReturnPrice()<250000)
            Vin.VIEW();
    }
    f.close();
}

```

}

5. (a) What is the difference between degree and cardinality of a table? What is the degree and cardinality of the following table? 2 Marks

Eno	Name	Salary
101	John Fedrick	45000
103	Raya Mazumdar	50600

Ans: Degree of a table signifies the number of columns or attributes of a table while Cardinality of a table signifies the number of rows or records in the table.

Degree of the given table = 3

Cardinality of the given table = 2

Note:

Write SQL queries for (b) to (g) and write the outputs for the SQL queries mentioned shown in (h1) to (h4) parts on the basis of tables ITEMS and TRADERS.

Table: ITEMS

CODE	INAME	QTY	PRICE	COMPANY	TCODE
1001	DIGITAL PAD 12i	120	11000	XENITA	T01
1006	LED SCREEN 40	70	38000	SANTORA	T02
1004	CAR GPS SYSTEM	50	21500	GEOKNOW	T01
1003	DIGITAL CAMERA 12X	160	8000	DIGICLICK	T02
1005	PEN DRIVE 32 GB	600	1200	STOREHOME	T03

Table : TRADERS

TCODE	TNAME	CITY
T01	ELECTRONIC SALES	MUMBAI
T03	BUSY STORE CORP	DELHI
T02	DISP HOUSE INC	CHENNAI

(b) To display the details of all the items in ascending order of product names(i.e. INAME). **1 Mark**

Ans: SELECT * FROM ITEMS WHERE ORDER BY INAME';

(c) To display item name and price of all those items, whose price is in the range of 10000 and 22000 (both value inclusive). **1 Mark**

Ans: SELECT INAME, PRICE FROM ITEMS WHERE PRICE BETWEEN 10000 AND 22000;

(d) To display the number of products, which are supplied supplier, i.e., the expected output should be: **1 Mark**

T01 2

T02 2

T03 1

Ans: SELECT TCODE, COUNT(TCODE) FROM ITEMS GROUP BY TCODE;

(e) To display the price, item name and quantity (i.e., qty) of those products which have quantity more than 150. **1 Mark**

Ans: SELECT PRICE, INAME, QTY FROM ITEMS WHERE QTY>150;

(f) To display the names of those traders, who are either from DELHI or from MUMBAI. **1 Mark**

Ans: SELECT TNAME FROM TRADERS WHERE CITY IN ("DELHI", "MUMBAI").

(g) To display the name of the companies and the name of the items in descending order of company names. 1 Mark

Ans: SELECT COMPANY, INAME FROM ITEMS ORDER BY COMPANY DESC;

(h) Obtain the outputs of the following SQL, queries based on the data given in tables ITEMS and TRADERS above. 2 Marks

h1) SELECT MAX(PRICE), MIN(PRICE) FROM ITEMS;

Ans: MAX(PRICE) MIN(PRICE)
38000 1200

h2) SELECT PRICE*QTY AMOUNT FROM ITEMS WHERE CODE=1004;

Ans: AMOUNT
1075000

h3) SELECT DISTINCT TCODE FROM ITEMS;

Ans: DISTINCT TCODE
3

h4) SELECT INAME, TNAME FROM ITEMS I, TRADERS T WHERE I.TCODE = T.TCODE AND QTY<100;

Ans: INAME TNAME
LED SCREEN 40 DISP HOUSE INC
CAR GPS SYSTEM ELECTRONIC SALES

6. (a) Verify the following using Boolean Laws 2 Marks

$$A + C = A + A'.C + B.C$$

Ans: $A + A'.C + B.C$

$= (A + A')(A + C) + B.C$ (Distributive Law)

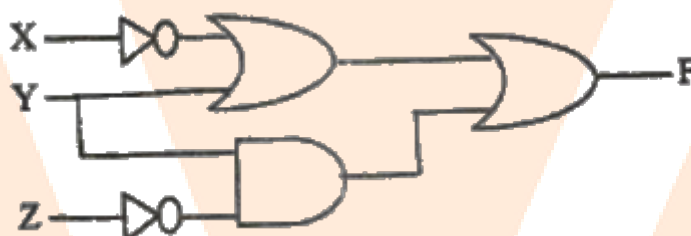
$= A + C + B.C$

$= A + C.(1 + B)$ (OR Absorption Law)

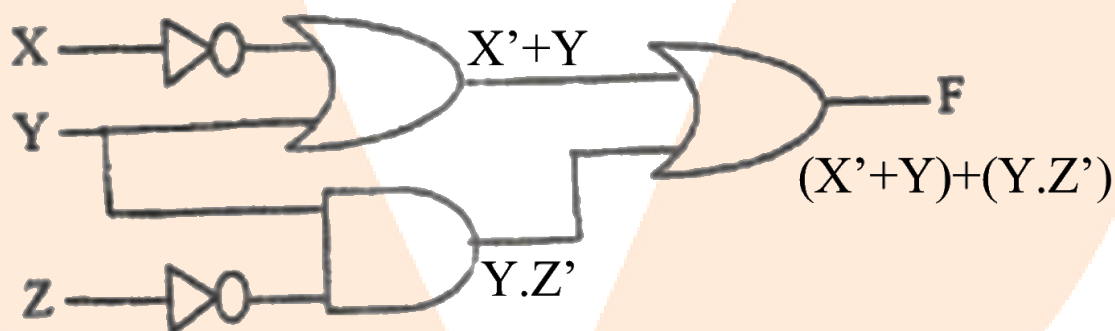
$= A + C$

(b) Obtain the Boolean Expression for the logic circuit shown below:

2 Marks



Ans: $(X' + Y) + (Y.Z') = F$



(c) Write the Sum of Product form of the function $G(U, V, W)$, for the following truth table representation of F .

1 Mark

X	Y	Z	G
0	0	0	1
0	0	1	0
0	1	0	1

0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

Ans: $G = (X'Y'Z') + (X'YZ') + (XY'Z') + (XYZ)$

$G = X'Z'(Y'+Y) + XY'Z' + XYZ$ (Complement Law)

$G = X'Z' + XY'Z' + XYZ$

$G = Z'(X'+XY') + XYZ$

$G = Z'(X'+Y') + XYZ$ (Redundancy Law)

$G = X'Z' + Y'Z' + XYZ$

(d) Obtain the minimal form for the following Boolean expression using map.
3 Marks

$H(P, Q, R, S) = \Sigma(0,1,2,3,5,7,8,9,10,14,15)$

Ans: K-map is as follows:

	R'S'	R'S	RS	RS'
P'Q'	1	1	1	1
P'Q	0	1	1	0
PQ	0	0	1	1
PQ'	1	1	0	1

Quadrant 1: P'Q'

Quadrant 2: P'S

Quadrant 3: Q'S'

Pair 1: PQ'R'

Pair 2: PQR

$$H = P'Q' + P'S + Q'S' + PQ'R' + PQR$$

7. (a) What is the difference between domain name and IP address? 1 Mark

Ans: IP address is the identifier for the computer on the network while Domain name is the DNS name which identifies the concerned IP address.

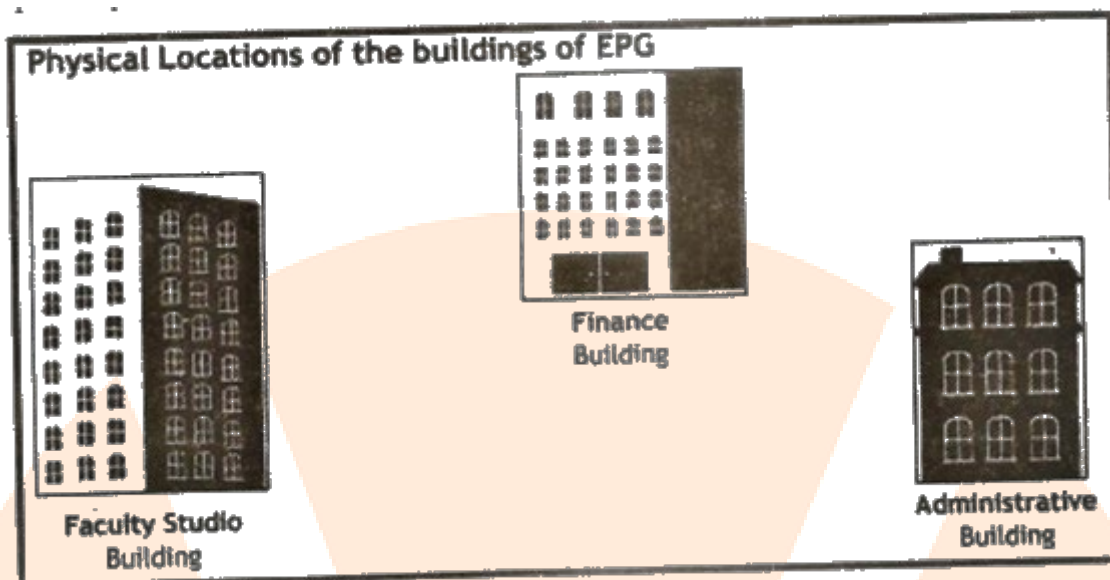
Example: 192.125.38.101 is the IP address while www.example.com is the domain name.

(b) Write two advantages of using an optical fibre cable over an Ethernet cable to connect two service stations, which are 190m away from each other. 1 Mark

Ans: Advantages of optical fibre over ethernet cable are:

- Optical fibre provides faster speed than ethernet cable.
- The data received at the other end will be less disturbed by noise.

(c) Expertia Professional Global (EPG) is an online corporate 4 training provider company for IT related courses. The company is setting up their new campus in Mumbai. You as a network expert have to study the physical locations of various blocks and the number of computers to be installed. In the planning phase, provide the best possible answer for the queries (i) to (iv) raised by them. 4 Marks



Building to Building Distances (in Mtrs)

From	To	Distance
Administrative Building	Finance Building	60
Administrative Building	Faculty Studio Building	120
Finance Building	Faculty Studio Building	70

Expected Computers to be installed in each Building :

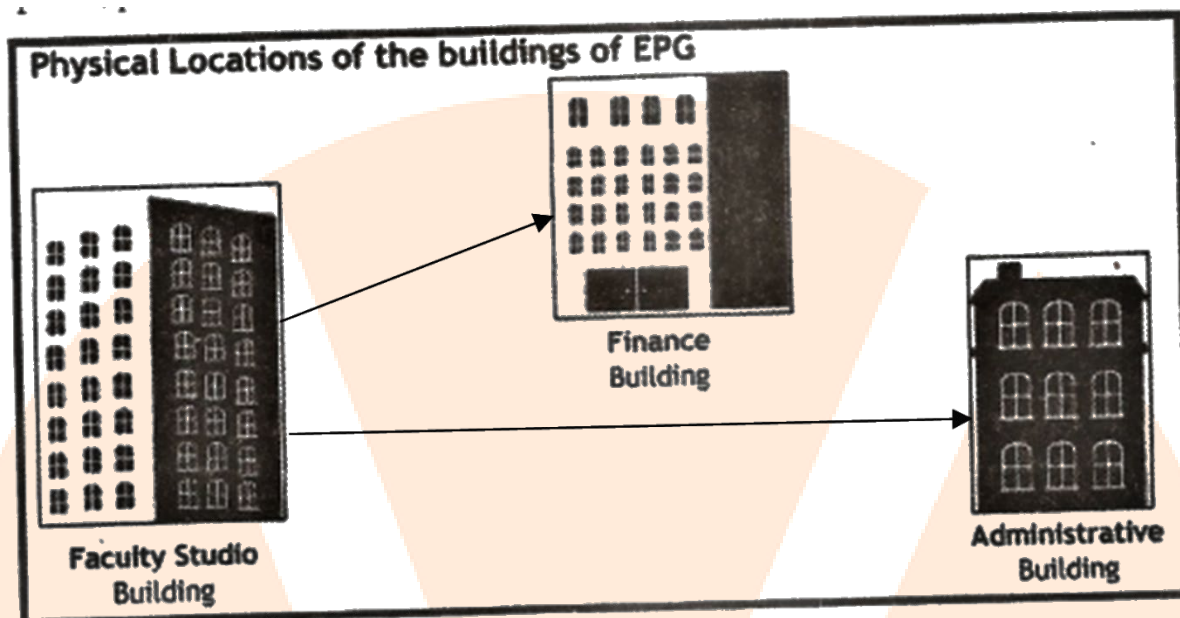
Buildings	Computers
Administrative Building	20
Finance Building	40
Faculty Studio Building	120

(i) Suggest the most appropriate building, where EPG should plan to install the server.

Ans: Faculty Studio Building to install the server.

(ii) Suggest the most appropriate building to building cable layout to connect all three buildings for efficient communication.

Ans:



(iii) Which type of network out of the following is formed by connecting the computers of these three buildings?

- LAN
- MAN
- WAN

Ans: LAN is formed connecting the three buildings.

(iv) Which wireless channel out of the following should be opted by EPG to connect to students from all over the world?

- Infrared
- Microwave Satellite

Ans: Microwave Satellite

(d) Write two advantages of using proprietary software over open source software. **1 Mark**

Ans: Advantage of proprietary software over open source software:

- Proprietary software can be easy to use as the user doesn't need to worry about development of the software.

- For open source software, there may be less support available to fix the bugs, the user has to depend upon the community of its users.

(e) Which of the following crime(s) is/are covered under cybercrime? 1 Mark

(i) Stealing brand new hard disk from a shop.

(ii) Copying some important data from a computer without taking permission from the owner of the data.

(iii) Getting into unknown person's social networking account and start messaging on his behalf.

Ans: (ii) Copying some important data from a computer without taking permission from the owner of the data.